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Cold-start active learning for image classification

Qiuye Jin^{a,b}, Mingzhi Yuan^{a,b}, Shiman Li^{a,b}, Haoran Wang^{a,b}, Manning Wang^{a,b,*},
Zhijian Song^{a,b,*}

^a Digital Medical Research Center, School of Basic Medical Sciences, Fudan University, Shanghai 200032, China^b Shanghai Key Laboratory of Medical Image Computing and Computer Assisted Intervention, Shanghai 200032, China

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ABSTRACT

Active learning (AL) aims to select valuable samples for labeling from an unlabeled sample pool to build a training dataset with minimal annotation cost. Traditional methods always require partially and initially labeled samples to start active selection and then query annotations of samples incrementally through several iterations. However, this scheme is not effective in the deep learning scenario. On the one hand, initially labeled sample sets are not always available in the beginning. On the other hand, the performance of the traditional model is usually poor in the early iterations due to limited training feedback. For the first time, we propose a cold-start AL model based on representative (CALR) sampling, which selects valuable samples without the need for an initial labeled set or the iterative feedback of the target models. Experiments on three image classification datasets, CIFAR-10, CIFAR-100 and Caltech-256, showed that CALR achieved a new state-of-the-art performance of AL in cold-start settings. Especially in low annotation budget conditions, our method can achieve up to a 10% performance increase compared to traditional methods. Furthermore, CALR can be combined with warm-start methods to improve the start-up efficiency while further breaking the performance ceiling of AL, which makes CALR have a broader application scenario.

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1. Introduction

Deep neural networks (DNNs) facilitate the utilization of large datasets through the direct learning of correlations between raw input data and target outputs, often requiring a large number of labeled samples for training to achieve a high level of performance [1]. However, in many applications such as medical imaging, the cost of annotation is very expensive. As a result, methods for reducing the annotation cost of DNNs while maintaining their performance are highly desirable [2]. Active learning (AL) is a promising solution to this problem, which selects valuable unlabeled samples for labeling to improve the performance of the trained model as much as possible under a limited annotation budget [3–5].

Unfortunately, traditional AL methods, such as uncertainty sampling [6–10] and committee-based algorithms [11,12], perform poorly in deep-learning scenarios. These methods work iteratively and incrementally to select samples for labeling, and the selection at each round depends on the trained target DNNs for calculating uncertainty or finding disagreements in predictions. However, the performances of DNNs depend heavily on the number of training data, but in the early rounds of iterations, the size of the training set is small. Therefore, the performances of trained DNNs tend to be low in the early

* Corresponding authors at: Digital Medical Research Center, School of Basic Medical Sciences, Fudan University, Shanghai 200032, China.

E-mail addresses: mnwang@fudan.edu.cn (M. Wang), zjsong@fudan.edu.cn (Z. Song).

rounds, and the calculations of uncertainties or disagreements are unreliable, which usually makes the sample selection inefficient and even worse than random selection [13–15]. To make traditional AL methods work in deep learning scenarios, a large number of samples with initial annotations are often needed for the initialization of AL models. In addition, in the iterative way of traditional AL methods, the sample selection process must be suspended several times until newly selected samples are annotated, which is not practical in many real-world applications [13].

To solve the problems mentioned above, Yuan et al. [16] proposed the conception of cold-start AL in the field of natural language processing (NLP), in which all the valuable samples are selected at one time without the need for an initial labeled set. In contrast, we name the traditional AL approaches warm-start approaches, which require initial annotations and iterative optimizations with multiple query requests for experts. As shown in Fig. 1, compared with warm-start AL, cold-start AL cannot make use of information from annotations or feedback from models, thus making sample selection more difficult. As a result, a completely different sampling strategy from that of warm-start AL needs to be designed.

To date, however, there has been no study to design cold-start AL models for image data. To the best of our knowledge, we propose for the first time a cold-start active learning framework based on representative (CALR) sampling in the field of computer vision that can select informative samples without any initially labeled samples. To achieve cold-start AL, we need to solve two pivotal issues: extracting informative features unsupervised and developing a criterion of valuable samples without feedback from the classification model. For the first issue, we use the method of advanced contrastive self-supervised learning to obtain a favorable representation of the feature space. For the second issue, we develop an effective data selection approach based on agglomerative hierarchical clustering [17]. Experimental results on the CIFAR-10, CIFAR-100, and Caltech-256 datasets show that our proposed method can conduct a one-shot selection of valuable samples without initial annotations and can outperform a series of comparison methods, including random sampling and even warm-start AL methods.

Our main contributions are as follows.

1. For the first time in the field of image classification, we propose a cold-start AL framework named CALR, which can select samples at one time without the need for an initial annotation set.
2. We propose a novel sample selection strategy based on hierarchical clustering and show that bottom-up hierarchical clustering is suitable for cold-start AL, especially in handling class-imbalanced datasets.
3. We conducted extensive experiments on three image classification datasets, and the proposed method outperformed random sampling and even warm-start AL methods.

2. Related works

2.1. Query strategy of AL methods

AL methods existed before deep learning became popular [3,18]. The goal of AL is to select valuable samples in the unlabeled sample pool as a core set and assign them to experts for annotations to reduce the annotation cost while ensuring the performance of the classification model at the same time. Therefore, developing a well-designed query strategy, i.e., a favorable criterion for selecting valuable samples, is the main task of AL methods.

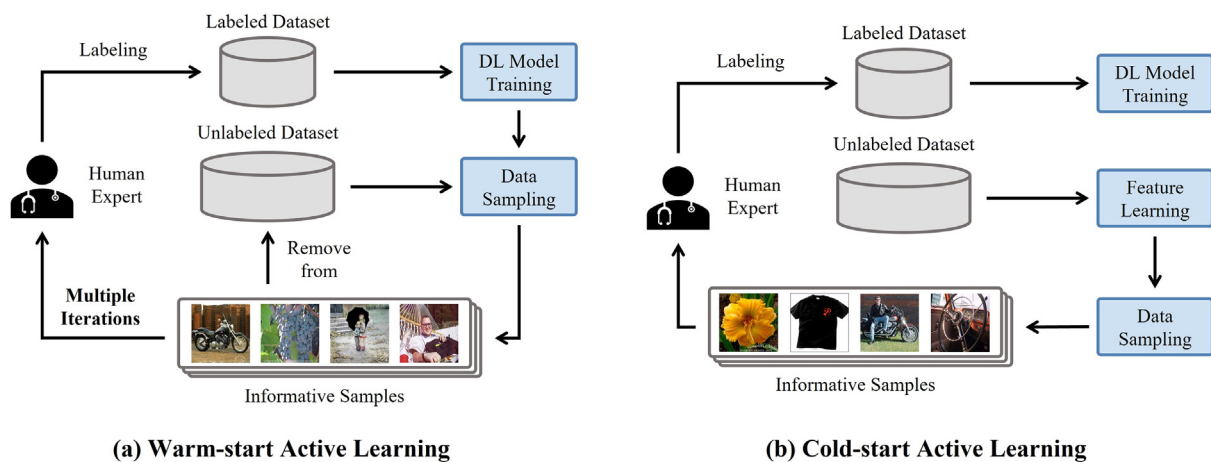


Fig. 1. Frameworks of warm-start active learning versus cold-start active learning. The warm-start setting allows pretraining with a partially annotated dataset and multiple interactions with the expert, allowing the model to be updated multiple times for informative feedback. In contrast, the cold start setting allows no initial annotated samples to be available and only one interaction with the expert.

There have been many ways of formulating such query strategies. Among them, the two most widely used are uncertainty sampling [6–8,10] and committee-based algorithms [11,12]. Uncertainty sampling assumes that the samples with higher uncertainty provide more information for the training of a classifier if they are labeled. Three metrics have been proposed to measure the uncertainty of samples, including least-margin [19], least confidence [6,7] and max entropy [8,20]. Because of its simplicity and low computational cost, the uncertainty-based approach has also been widely used in the era of deep learning [21–24]. Another popular query strategy is based on the disagreement of committees [11,12,25–27], which uses a collection of multiple classifiers to measure the divergence for each unlabeled sample and select the samples with the largest divergence scores. The underlying assumption of the committee-based algorithm is that if the predictions of classifiers for certain samples are inconsistent, querying the annotations for these samples will provide more information.

Other query strategies have been developed over the years, e.g., expected model change [28] and density weighted methods [14]. For more details, the review article [29] can be used for reference.

2.2. Initialization of warm-start AL methods

Although many classical AL models have been proposed, such as uncertainty-based methods and committee-based methods, they all fall into the category of the warm-start paradigm; that is, they need a certain number of labeled samples to achieve the initialization of the selection process. The quality of initialization directly affects the performance of warm-start AL [30].

In the early era of machine learning, only a small number of labeled samples or even a single labeled sample can be enough to perform the initialization of warm-start AL models because feature descriptors are manually extracted already, and only the inductive bias needs to be fitted. Therefore, warm-start AL methods commonly use density weighted methods [14] to determine the samples for initialization, namely, to select the samples with the highest information density [3]. In contrast, in the deep learning era, classifiers depend on end-to-end trainable deep neural networks, where feature extraction and inductive bias are integrated into one unified framework. In this context, the initialization of warm-start AL methods cannot be completed with a single sample or a small number of samples.

Due to the lack of an effective approach, most warm-start deep AL methods use random initialization, and there are only a few studies that focus on the initializations of deep AL methods. In the field of medical image object detection, Zhou et al. [15] first used transfer learning to extract features and then utilized an ensemble model including randomness, uncertainty and majority to query the sample set for the initialization of warm-start AL methods. However, the main challenge for their method is the large domain shift between the source domain ImageNet and the target domain of medical images, which substantially affects initialization.

2.3. Cold start active learning

Cold-start AL is a low-budget, low-resource paradigm of AL that only allows one opportunity to query annotations from experts and cannot utilize initially annotated data (including validation data) or update the active selection model [26]. Cold-start AL is more difficult than the usual warm-start AL because it requests a one-shot selection of samples for annotation and cannot be optimized using the feedback of the prediction result. In the cold-start setting, traditional sample query strategies are impractical because of model instability and data scarcity [16].

To date, studies on cold-start AL are very limited. In the field of natural language processing (NLP), Yuan et al. [16] used a pretrained BERT model for feature extraction and then used the loss of language modeling to select samples that ‘surprised’ the AL model. In the field of medical image processing, Zheng et al. [13,31] proposed a representation-based one-shot AL framework, which was composed of a variational autoencoder for feature extraction and a sampling module based on k-means and max-cover algorithms. Both of these methods mentioned above can be regarded as cold-start approaches. However, there is no suitable cold-start AL framework for natural image classification. Accordingly, to fill the current research gap on natural image classification, we propose a novel cold-start AL method based on contrastive learning and hierarchical clustering.

3. Method

3.1. Problem definition

We assume that $U = \{x_1, x_2, x_3, \dots, x_n\}$ is an unlabeled dataset containing n samples, and the goal of AL is to select a core subset $C \subset U$ that includes m samples with the highest annotation value. Then, we ask for labels of C from an oracle O (human experts), and finally, we can obtain a labeled dataset L for training a classifier f . For deep active learning (DAL), the classifier f is a deep neural network.

In general, the selection of C involves T iterations. For warm-start AL, the total iterations are $T > 1$, and they need to suspend the selection process for T times to expand the training set and update the classifier iteratively. For cold-start AL, the total iterations are $T = 1$, and they can only ask for annotations from oracles once, with no iterative updating of the classifier. Before starting the selection, warm-start AL methods need an initial labeled dataset L_0 for training to obtain an initial clas-

sifier f_{θ_0} . After the initialization of the classifier, warm-start AL methods use a query strategy S , which needs the feedback from the classifier f in classifying unlabeled samples, to iteratively select unlabeled samples and add them to C for retraining and updating the classifier. In contrast, cold-start AL methods do not need an initial labeled dataset for model initialization, and its query strategy S does not need feedback from the classifier. Cold-start AL uses its query strategy to select unlabeled samples to form C at one time.

The key component of AL is the query strategy S because it determines which samples are label-efficient and then selects them to form the core set C . The query strategy S is implemented by a query model M_t . In the setting of warm-start AL, the query model M_t is the classifier $f_{\theta_{t-1}}$ with the weight θ_{t-1} after $t - 1$ iterative updates, where θ is the parameter of the classifier. Under the setting of cold-start AL, the query is performed only once, and the query model M_1 does not depend on the classifier. The query model for cold-start AL can only select valuable samples by exploring the unlabeled samples themselves.

3.2. Key components

To effectively select valuable samples under the condition of a cold start, we designed a representation-based AL method named CALR. This method consists of three key components (Fig. 2): (a) feature extraction, through mapping images into the latent space (feature space) by contrastive self-supervised learning; (b) clustering, using the BIRCH algorithm to cluster the feature vectors into m clusters; and (c) representative selection, by selecting the most representative sample from each cluster to form the core set C . Afterward, C is given to the experts, and the labeled set L can be obtained. In the following subsections, we will explain these three components in detail.

3.3. Contrastive self-supervised feature extraction

The quality of feature representation has a great influence on the effectiveness of active learning [32]. For cold-start active learning, we do not have initially labeled samples to learn feature representation, so we propose utilizing a contrastive self-supervised learning algorithm.

Contrastive self-supervised learning is the most advanced unsupervised feature representation model. The idea is to maximize the consistency between similar samples (positive sample pairs) and minimize the consistency between dissimilar samples (negative sample pairs). To achieve this goal, we use an encoder network to extract features and then evaluate the similarity of the sample pairs in the feature space using a similarity score. In this process, the encoder can learn a good representation of images through a “learning comparison” in the feature space [33]. Specifically, we use a series of data augmentations on the same image to construct positive sample pairs and use different images to construct negative sample pairs. With these positive and negative sample pairs, we use the noise contrastive estimation [34] (NCE) objective function to “learn comparison”:

$$\mathcal{L}_0 = \mathbb{E}_{x, x^+, x^-} \left[-\log \left(\frac{e^{f(x)^T f(x^+)}}{e^{f(x)^T f(x^+)} + \sum_{k=1}^K e^{f(x)^T f(x^-)}} \right) \right]$$

where x^+ is similar to x as a positive sample, x^- is different from x as a negative sample, f is an encoder to extract the features from the original image, and K is the number of negative samples.

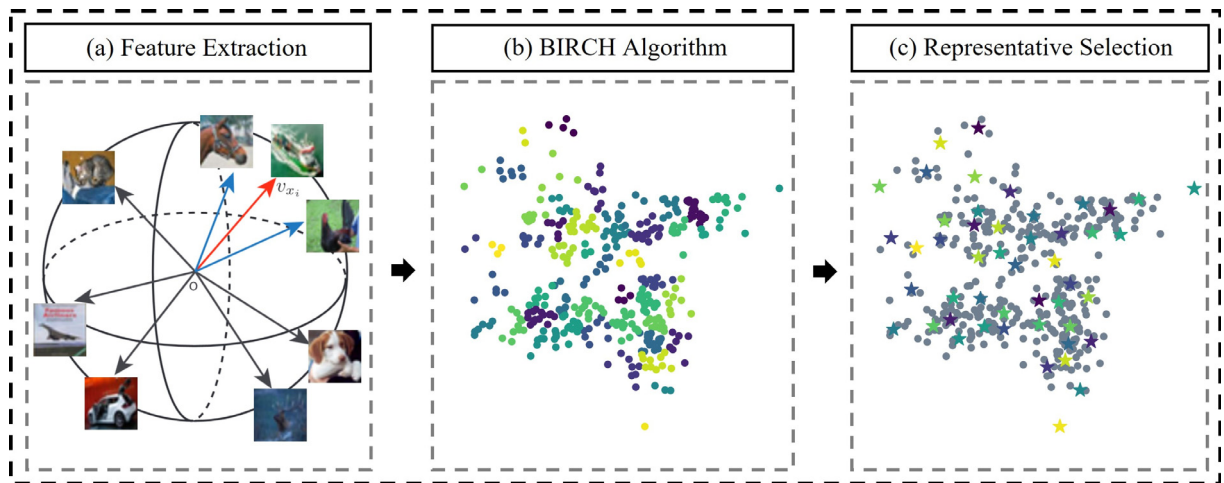


Fig. 2. Key components of CALR: (a) contrastive self-supervised feature extraction; (b) BIRCH clustering for the feature space; (c) selection of samples based on the representativeness, and the pentagrams represent the selected samples.

We further use the momentum contrast paradigm [35] to construct a self-supervised model; that is, a momentum encoder is used as a dynamic dictionary for encoding negative samples during training. In this way, by using two encoders, namely, the query encoder $f_q(\cdot)$ and momentum key encoder $f_k(\cdot)$, the fluctuation of the loss convergence in the initial training stage can be suppressed. For example, given an input image x , we want to learn the original representation $q = f_q(x)$ by the query encoder, which can distinguish x from other images. Meanwhile, for a set of other images x_i , we use an asynchronously updated momentum key encoder to encode positive and negative samples and optimize the following objective function:

$$\mathcal{L} = -\log \frac{\exp(q \cdot k_+/\tau)}{\sum_{i=0}^K \exp(q \cdot k_-/\tau)}$$

where τ is a hyperparameter temperature, k_+ is the encoding of the momentum key encoder to the positive sample $f_k(x^+)$, and k_- is the encoding of the momentum key encoder to the negative sample $f_k(x^-)$. To expand the capacity and sampling efficiency of constructing sample pairs, the output of the momentum encoder is stored by a queue, which adds the latest encoded batch to the end of the queue while deleting the old coding features at the head of the queue. In addition, the momentum encoder and the query encoder use the same network architecture. The momentum encoder does not need to update parameters immediately after each forward transmission but updates parameters according to the query encoder:

$$\theta_k \leftarrow c\theta_k + (1 - c)\theta_q$$

where $c \in [0, 1)$ is the momentum coefficient; and θ_k and θ_q are the weights of the momentum encoder and query encoder, respectively. Only θ_q is updated through backpropagation, and the existence of the momentum coefficient makes θ_k update more smoothly.

Once the training of the feature extractor is finished, we can transform the image into a set of d -dimensional features: $Z = \{z_1, z_2, \dots, z_i, \dots, z_n\}$. Next, we are able to select samples in the feature space through our designed query model $M_1(\cdot)$.

3.4. Clustering-based query model

We present the data selection approach in Algorithm 1, in which we first extract the feature for each image and use BIRCH clustering to obtain m clusters. Each cluster may contain n_c samples. Moreover, we find the sample with the highest information density as the most representative sample of the cluster, and it is selected to be annotated. Then, we decode the sampling result in the feature space back to the image space and obtain the annotated sample set X_L .

Algorithm 1: Cold-start active learning by BIRCH clustering and representative sampling

Input: unlabeled dataset U , feature extractor $f(\cdot)$, number of clusters m

Output: core set sample set X_C , annotated sample set X_L

- 1: Use the pretrained encoder $f(\cdot)$ to extract feature bank Z :
 $f(U) \rightarrow Z = \{z_1, z_2, \dots, z_n\}$;
 - 2: Group the feature set Z into m clusters via BIRCH clustering:
 $C^* = \{C_i^* | i = 1, 2, \dots, m\}$,
 $C_i^* = \{z_{ij} | j = 1, \dots, n_c\}$;
 - 3: Select the samples with the highest information density in each cluster:
 $z_i^* = \underset{z_{ij} \in C_i^*}{\operatorname{argmax}} I(z_{ij})$;
 $C = \{z_i^* | i = 1, 2, \dots, m\}$;
 - 4: Decode the sampling results in feature space back to image space:
 $X_C = \{x_i | i = 1, 2, \dots, m\}$;
 - 5: Submit the annotation request to O for X_C :
 $X_C \xrightarrow{O} X_L$;
 - 6: **Return** X_L
-

3.4.1. BIRCH clustering.

In this study, the BIRCH algorithm [17] was used for clustering. BIRCH stands for the balanced iterative reducing and clustering using hierarchies, which is computationally efficient and suitable for large amounts of data and large numbers of categories [17].

The BIRCH algorithm is a kind of bottom-up hierarchical clustering, and its main idea is to use cluster features (CFs) to represent a cluster and use cluster feature trees (CF-trees) to represent the whole clustering hierarchy. CF is a triplet, i.e., $CF = (N, LS, SS)$, where N is a scalar, representing the number of samples in the cluster; LS is a d -dimensional vector, which is the linear sum of N samples in the cluster, i.e., $LS = z_1 + z_2 + \dots + z_N$; and SS is a scalar, which is the sum of the squared

lengths of the N samples in the cluster, i.e., $SS = |z_1|^2 + |z_2|^2 + \dots + |z_N|^2$. The above attributes contained in the CF can be updated when the representing cluster splits or merges, and they can be used to calculate some important properties of the cluster. For example, LS/N is the centroid of the cluster.

The CF-tree is the core module of the BIRCH clustering algorithm. There are several parameters constraining the structure of a CF tree. First, any non-leaf node can contain several child nodes, but the number of child nodes must be less than a threshold B . Second, the leaf node of a CF tree is composed of several CFs, but the number of CFs must be less than a threshold L . The attribute SS of a CF can approximately represent the diameter of a cluster, which must be less than a threshold T . To better describe the clustering process, we define the distance between a sample z_i and a cluster C_i as:

$$d(z_i, C_i) = \sqrt{(z_i - LS_i/N_i)^2}$$

where N_i is the N attribute in the CF corresponding to the cluster C_i , and LS_i is the LS attribute in the CF corresponding to the cluster C_i . Then, we define the distance between a sample z_i and a leaf node L_i :

$$d(z_i, L_i) = \sqrt{(z_i - \bar{LS}_i)^2}$$

where,

$$\bar{LS}_i = \frac{\sum_{i=0}^{b-1} LS_i}{\sum_{i=0}^{b-1} N_i}$$

Furthermore, we define the distance between two clusters C_i and C_j :

$$d(C_i, C_j) = \sqrt{(LS_i/N_i - LS_j/N_j)^2}$$

where N_j is the N attribute in the CF corresponding to the cluster C_j , and LS_j is the LS attribute in CF_j .

The process of BIRCH clustering is essentially the process of constructing a CF tree. The CF-tree is empty at the beginning. We first randomly select a sample from the set Z and put it into the root node. Then, we continuously select new samples from Z and insert them one by one into the CF-tree. The clustering process is shown in Algorithm 2: Because the number of clusters is determined as m in advance, we first construct a CF tree with a total number of clusters that are no less than m . Specifically, we can set the maximum number of subclusters $L = \lfloor \frac{n}{m} \rfloor$. This insertion process only scans the dataset once, which has a high computational efficiency and linear $O(n)$ time complexity. After the CF-tree is constructed, the clustering results are also determined. Then, we calculate the distance between two clusters in each leaf node. According to the principle of similarity merging, the two nearest clusters are merged into one cluster. After updating the distance, we merge again until the total number of clusters is m . In this way, we can divide the feature set $Z = \{z_1, z_2, \dots, z_i, \dots, z_n\}$ into m disjoint clusters, and then we can select the most representative sample from each cluster to form the final core set.

3.4.2. Representative sampling

The main idea behind the maximum density sampling is that the sample with the most information should be the sample that can best represent the potential distribution of a cluster, which is the sample located in the densest area of the latent space [36]. Therefore, we calculate the information density for each sample in a cluster and one with the largest information density from each cluster. Concretely, for the feature set $Z_c = \{z_1, z_2, \dots, z_j\}$ in a cluster, the information density of a single sample z can be defined as:

$$I(z) = \frac{1}{|Z_c|} \sum_{z' \in Z_c} \text{sim}(z, z')$$

where $\text{sim}(z, z')$ is a similarity function, and in this work, we use the cosine similarity:

$$\text{sim}(z, z') = \frac{z \cdot z'}{|z||z'|}$$

The higher the information density is, the more similar a given sample is to the rest of the data. Here, calculating the information density of one sample requires a time complexity of $O(\frac{n(n-1)}{2})$, where n represents the size of the feature set. Generally, the samples in the middle of the cluster have higher information densities; that is, they can better represent the whole set. Thus, instead of obtaining globally from the overall data scale n , we calculate the information density inside the clusters. At this time, the cluster data scale n_c is much smaller and reduces the computational complexity greatly. After calculating the information density of all samples in a cluster, we select the one with the highest information density.

4. Experiment

To evaluate the performance of CALR and compare it with existing AL methods, we conducted experiments on three public datasets: CIFAR-10 [37], CIFAR-100 [37], and Caltech-256 [38].

4.1. Implementation details

All DNNs in this study are implemented using PyTorch [39]. One NVIDIA GeForce RTX 3090Ti GPU and a 16 GB video memory are used for training and testing. The experimental setting of the contrastive self-supervised model is the same as that of He et al. [35]. Furthermore, we use its feature extractor as the backbone of our image classification network, and a fully connected layer is added behind the feature extractor to achieve image classifications. Some differences in the hyperparameters on different datasets will be described in detail in later sections.

We use sklearn [40] to implement the clustering algorithms used in all the AL methods. The Core-i9 processor and 64 GB of memory are used for computing and storage. More specifically, we implement the BIRCH algorithm with sklearn.cluster.Birch(.). The original BIRCH algorithm has two parameters, the branching factor and the threshold. The branching factor limits the number of subclusters in a node, and the threshold limits the distance between the entering sample and the existing subclusters. Here, the threshold is set to 0.5, and the branching factor is set to 50, i.e., both use the default values in sklearn. Furthermore, this algorithm is regarded as a data reduction method since it reduces the input data to a set of subclusters that are obtained directly from the leaves of the CF-tree. These reduced data can be further processed by inputting them into a global cluster. This global cluster is set by n_clusters. Here, n_clusters is equal to m , i.e., the number of queries. Then, a global clustering step labels these subclusters into m global clusters (pseudo-labels), and the samples are mapped to the global label of the nearest subcluster.

In addition, we use modAL [41] to calculate the information density of the samples. modAL is an open-source computing tool based on Python 3 and scikit-learn.

4.2. Competitors

We comprehensively compare the performance of **CALR** with different sample selection methods. First, we took **Random Sampling** as a baseline, which was shown to be rather competitive in existing research [42], and used the average of six runs with random seeds as its final performance score. For the advanced cold-start AL method **ALPS** [16] in NLP, we replace the BERT pretraining model with the contrastive self-supervised pretraining model for use on images. Its parameters and training process are consistent with our **CALR** to ensure fair comparison. The one-shot AL method **RA** [13] also works without an initial sample set, and we reproduce **RA** according to [13] and add it in the comparison. In addition, we also include the initialization method of warm-start AL **ACFT** [15] as a competitor of our cold-start AL.

At the same time, to show that our model is more competent for AL tasks under the condition of low resources and low budget, we include **Uncertainty** [10], the most popular model in the field of warm-start AL, as a competitor. Since **Uncertainty** belongs to the hot start model, we initialize it with random sampling.

To ensure the fairness of comparison, all competitors use the same unlabeled sample pool, and all classification networks and parameter settings are the same. In addition, all competitors repeat training six times and take the average to eliminate randomness. In addition, all competitors maintain the same training and test settings. The warm-start method **Uncertainty** allows multiple queries of samples and iterative updates of models, while the other cold-start methods can only query the sample and train the classification model once.

4.3. CIFAR-10 dataset

CIFAR-10 contains 10 categories of natural color images, including a total of 50,000 training images and 10,000 test images, and the image size is 32×32 . We compare the performance of **CALR** with other competitors and use the top-1 accuracy as the evaluation metric. Specifically, the number of training epochs is 200, the number of iterations is 150, the basic learning rate is 0.01, and the batch size is 256. A random gradient descent optimizer and cosine annealing learning schedule are used. Furthermore, we set the momentum to 0.9, and the learning rate weight decay to $5e-4$.

Fig. 3 shows the test results when the segmentation model was trained with different AL strategies. The **CALR** method achieves the best performance under different annotation budgets. Under the condition of a small annotation budget, the performance of warm-start model **Uncertainty** is limited or even worse than **Random** sampling. When the labeling budget increases, the performance of **Uncertainty** improves and gradually surpasses **Random** sampling. In contrast, the one-shot AL method **RA** performs well when the annotation budget is small, but it begins to lag the **Uncertainty** method when the annotation budget increases. The performance of several other groups of competitors is similar and cannot surpass **Random** sampling significantly.

4.4. CIFAR-100 dataset

CIFAR-100 is a natural color dataset similar to the CIFAR-10 dataset, and it contains 50,000 training images and 10,000 test images, with an image size of 32×32 , and the number of categories is 100. Specifically, we set the number of training epochs to 300, the number of iterations to 150, the basic learning rate to 0.01, and the batch size to 256. Moreover, the random gradient descent optimizer and cosine annealing learning schedule are used. Furthermore, we set the momentum to 0.9, and the learning rate weight decay to $5e-4$.

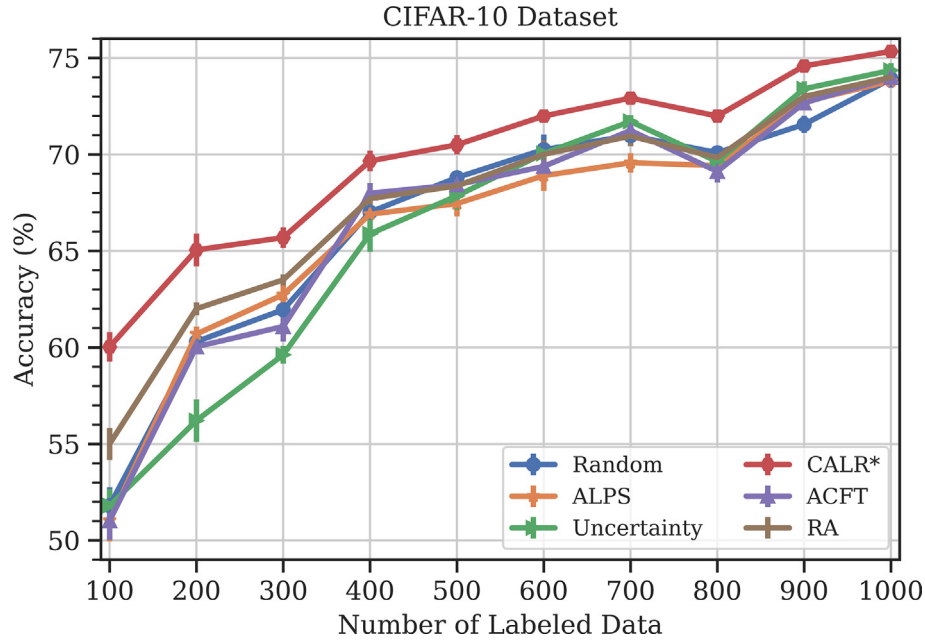


Fig. 3. Performance of different sampling methods on the CIFAR-10 dataset. * refers to our proposed method.

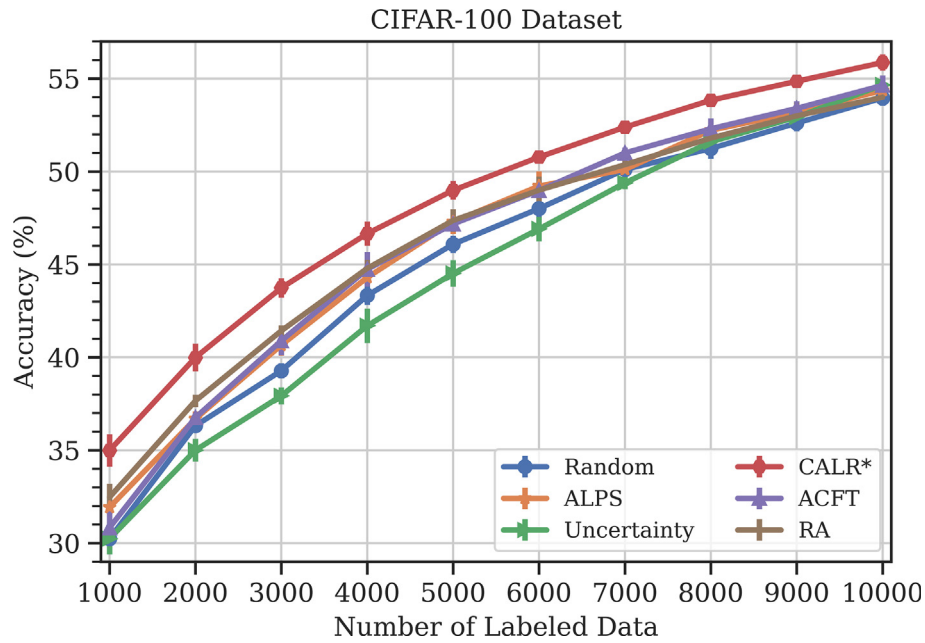


Fig. 4. Performance of different sampling methods on the CIFAR-100 dataset. * refers to our proposed method.

We show the performance of *CALR* and other competitors in Fig. 4, and we use the top-1 classification accuracy to indicate the performance. Experimental results show that *CALR* achieves the best performance under various annotation budgets. In this dataset, the performance of *RA*, *ALPS* and *ACFT* are similar, and they are all better than *Random* sampling. Similarly, the warm-start method *Uncertainty* performs worse than *Random* sampling when the annotation budget is small. When the annotation budget increases, the performance of *Uncertainty* improves gradually and eventually surpasses *Random* sampling.

4.5. Caltech-256 dataset

Caltech-256 is composed of 30,607 images in 256 categories. The number of images in each category is at least 80 and at most 827, and we utilize this dataset to evaluate the performance of all the methods when the classes are imbalanced. Caltech-256 only provides the annotation of the training set. To test the performance, we randomly select 10 images from each class to form the test set (2560 images in total), and the remaining images are used as an unlabeled sample pool for different sampling methods.

Specifically, we set the number of training epochs to 50, the number of iterations to 100, the basic learning rate to 0.001, and the batch size to 64. Moreover, the random gradient descent optimizer and cosine annealing learning schedule are used. Furthermore, we set the momentum to 0.9, and the learning rate weight decay to $5e-4$.

We show the performance of CALR and other competitors in Fig. 5, and we use the multiclassification accuracy to indicate the performance. The experimental results show that CALR can achieve the best performance under different annotation budgets. The performance of ALPS and RA is inferior to that of our proposed method and superior to that of the Random method. The performance of ACFT is close to that of Random sampling. In addition, the warm-start method Uncertainty performs worse than Random sampling under the condition of a low budget. When the annotation budget increases, it will gradually surpass Random sampling.

By comparing Fig. 5 with Figs. 3 and 4, we can see that the advantage of CALR is more significant when the classes are imbalanced. For the visualization result, we show the first five selected samples of different cold-start sampling methods in Fig. 6. As shown in Fig. 6, the most dominant classes, i.e., airplanes and mobiles, are shown in red boxes. The samples selected by CALR are the most category-balanced, i.e., we can avoid oversampling for the dominant classes.

4.6. Initialization of warm-start AL methods

Cold-start AL methods can not only achieve one-shot sampling under the condition of low resources but can also be used as the initialization method of warm-start AL methods. Therefore, we compared the performance of CALR and that of the competitors ACFT, ALPS and RA when they were used as the initialization methods of warm-start AL, Uncertainty [10]. The performance of initialization can be evaluated by the final performance of the classifier trained by samples selected both in the initialization and in the following iterative rounds of Uncertainty. The number of new queries in the iterative rounds is equal to the number of queries in the initialization (cold-start) stage.

We take Random initialization as the baseline (100 %) and use the ratio of the top-1 accuracy of different initialization methods to the accuracy of Random initialization as the score to measure the performance. As shown in Fig. 7, as the initialization method of warm-start AL, CALR performs best on all three datasets, significantly surpassing the other initialization methods.

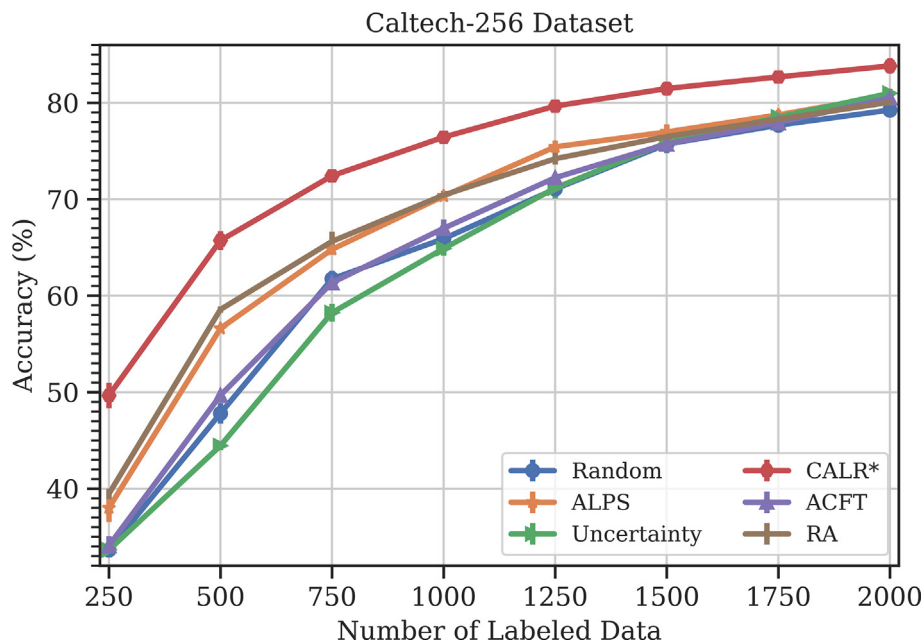


Fig. 5. Performance of different sampling methods on the Caltech-256 dataset. * refers to our proposed method.

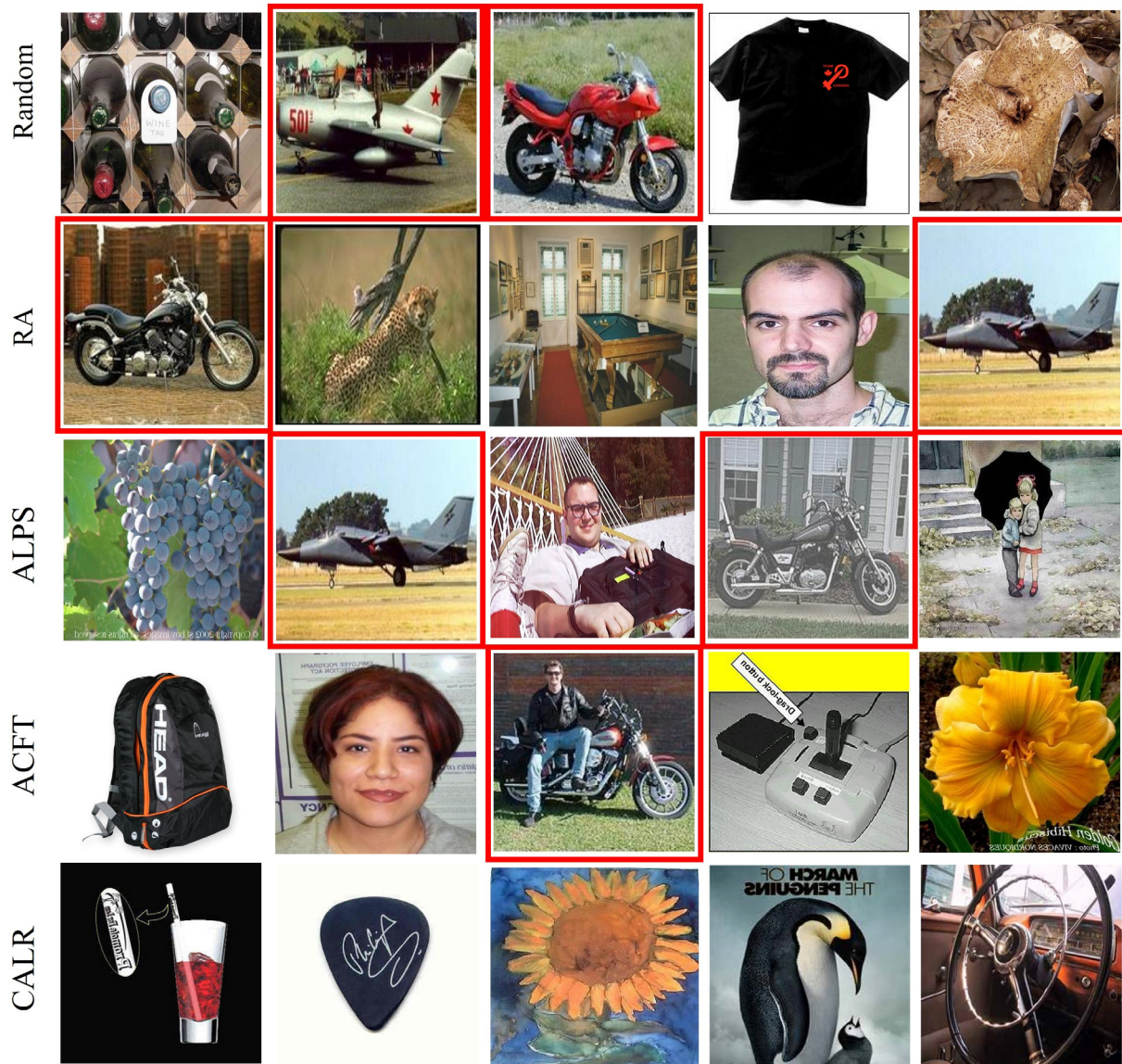


Fig. 6. First five query results for Caltech256 – *Random* (top), *RA* (second row), *ALPS* (third row), *ACFT* (fourth row) and our method (bottom). The most dominant classes are shown in red boxes.

In addition, we further compared the performance of CALR and that of the competitors *ACFT*, *ALPS* and *RA* when they were used as the initialization methods of warm-start AL, **Committee** [11], which is a relatively popular warm-start AL method. As shown in Fig. 8, CALR also performs best on all three datasets, significantly surpassing the other initialization methods. These results show that CALR can be used as an initialization method for various warm-start AL models.

4.7. Ablation study

To further analyze our proposed CALR method, we carried out several ablation tests. For feature extraction, we replace contrastive self-supervised pretraining with ImageNet pretraining and named it *CALR_IN*. For the clustering module, we replace the BIRCH algorithm with K-Means clustering and name it *CALR_KM*. In addition, we replace the above two components at the same time, i.e., the DEgraded CALR version of ImageNet pretraining + K-means clustering, and name it *CALR_DE*.

The ablation experiments on the CIFAR-10, CIFAR-100, and Caltech256 datasets are shown in Figs. 9–11. For the CIFAR datasets, *CALR_KM*, which uses K-means clustering, can achieve performance second only to CALR. *CALR_IN* is slightly worse than *CALR_KM*. The fully degraded version *CALR_DE* has the worst performance.

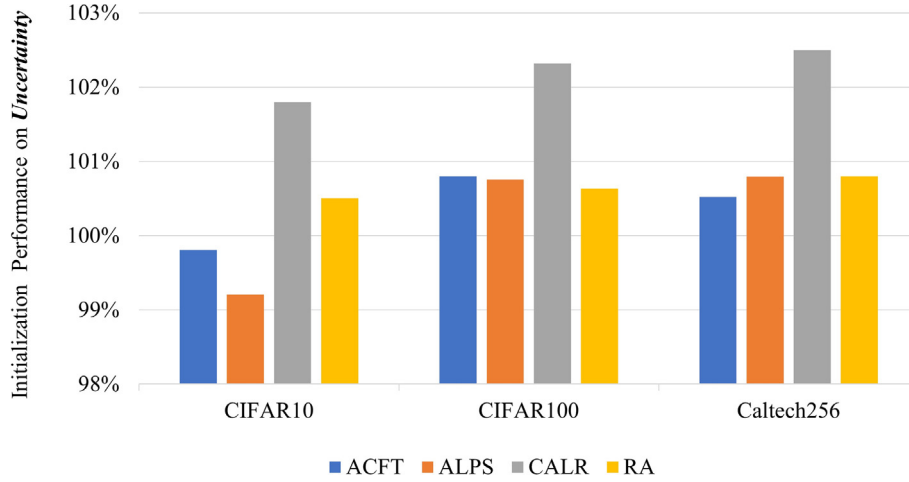


Fig. 7. Comparison of different initialization methods on the warm-start model *Uncertainty*. We take *Random* initialization as the baseline (100%) and take the ratio of the accuracy between any other initialization method and random initialization as the performance score.

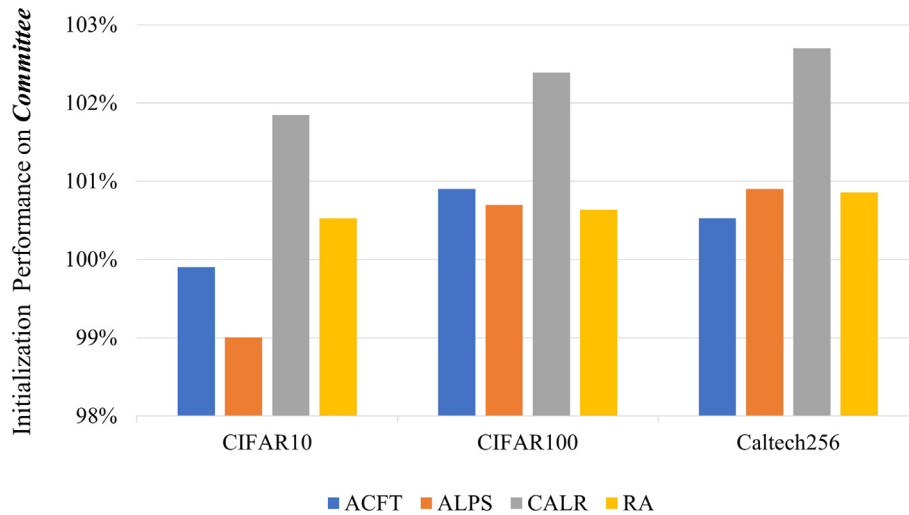


Fig. 8. Comparison of different initialization methods on the warm-start model *Committee*. We take *Random* initialization as the baseline (100%) and take the ratio of the accuracy between any other initialization method and random initialization as the performance score.

For the Caltech256 dataset, different from the CIFAR datasets, ImageNet pretrained *CALR_IN* can achieve performance second only to *CALR*. *CALR_KM*, which uses K-means clustering, is worse than *CALR_IN*. Similarly, the fully degraded version *CALR_DE* has the worst performance.

4.8. Comparative study

First, we conduct comparative experiments to further analyze the effect of the feature extractors on *CALR*. Specifically, we use the variational autoencoder [43], deep convolutional generative adversarial networks [44], rotation prediction [45], and contrastive learning [35] as feature extractors for *CALR*, and they are denoted as *CALR_VAE*, *VAE_DCGAN*, *CALR_ROT*, and *CALR*, respectively. It is notable that *CALR* is the original version of our proposed method.

The results of these comparative studies on the CIFAR-10, CIFAR-100, and Caltech-256 datasets are shown in Figs. 12–14, respectively. *CALR*, the original version using contrastive self-supervised learning for feature representation, performs best, while the performance of *CALR_ROT* is the second best. In addition, *CALR_DCGAN*, a model using a deep convolutional generative adversarial network, and *CALR_VAE*, a model using a variational autoencoder, have similar performance on these three datasets.

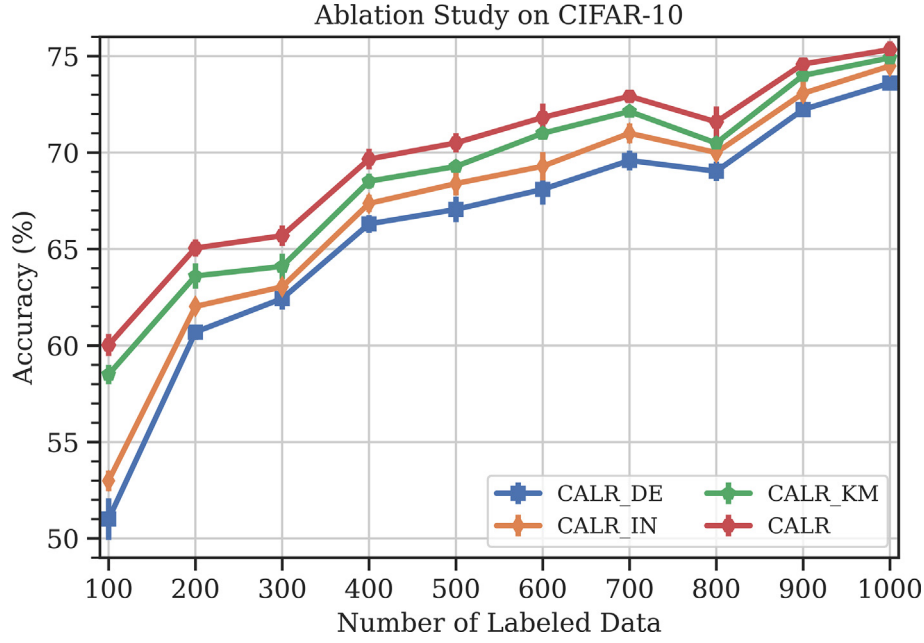


Fig. 9. Ablation experiments on the CIFAR-10 dataset.

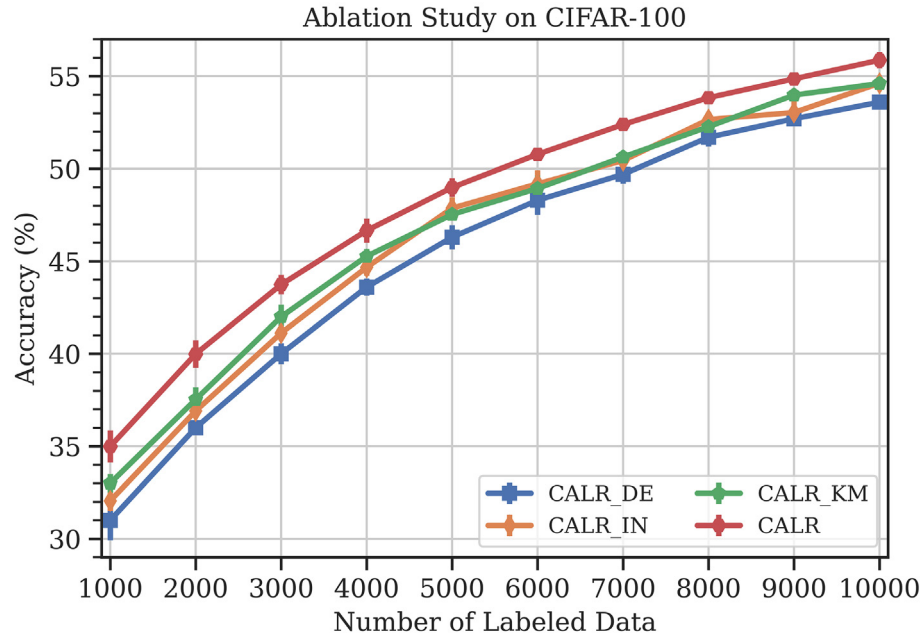


Fig. 10. Ablation experiments on the CIFAR-100 dataset.

Second, to further analyze the effect of the selection methods on CALR, we conducted relative comparative experiments. More specifically, after we calculate the information density, we select samples of different quartiles, i.e., Q_0 , Q_1 , Q_2 , Q_3 and Q_4 , which correspond to the 0th, 25th, 50th, 70th and 100th percentiles of the sample in ascending order of the information density. Obviously, Q_4 corresponds to the naïve method we proposed, i.e., the strategy of picking the maximum information density. In addition, we also added the comparison of clustering first and then randomly selecting points instead of ranking the information density. The methods mentioned above are noted as $CALR_{Q0}$, $CALR_{Q1}$, $CALR_{Q2}$, $CALR_{Q3}$, $CALR_{Q4}$ and $CALR_{RAND}$.

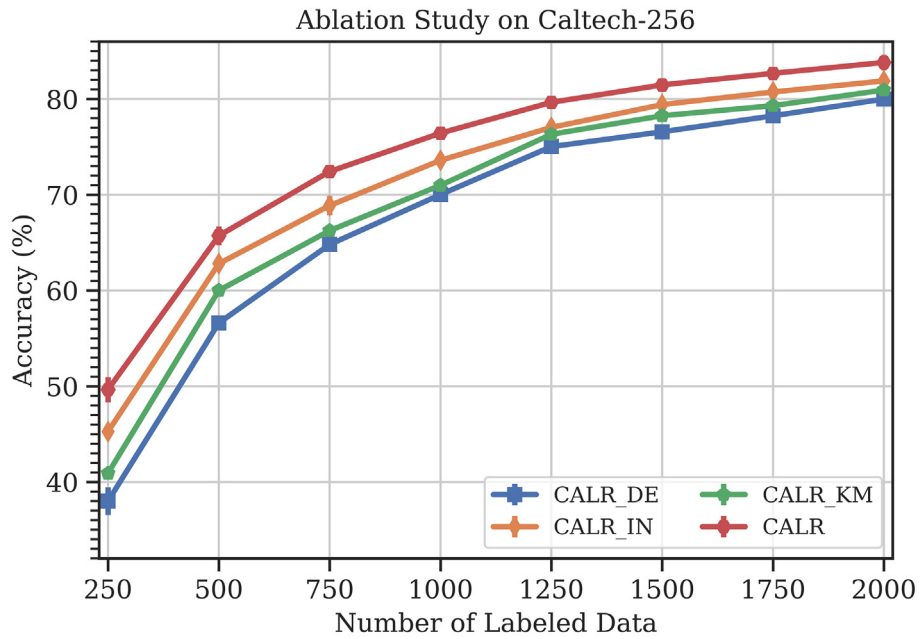


Fig. 11. Ablation experiments on the Caltech-256 dataset.

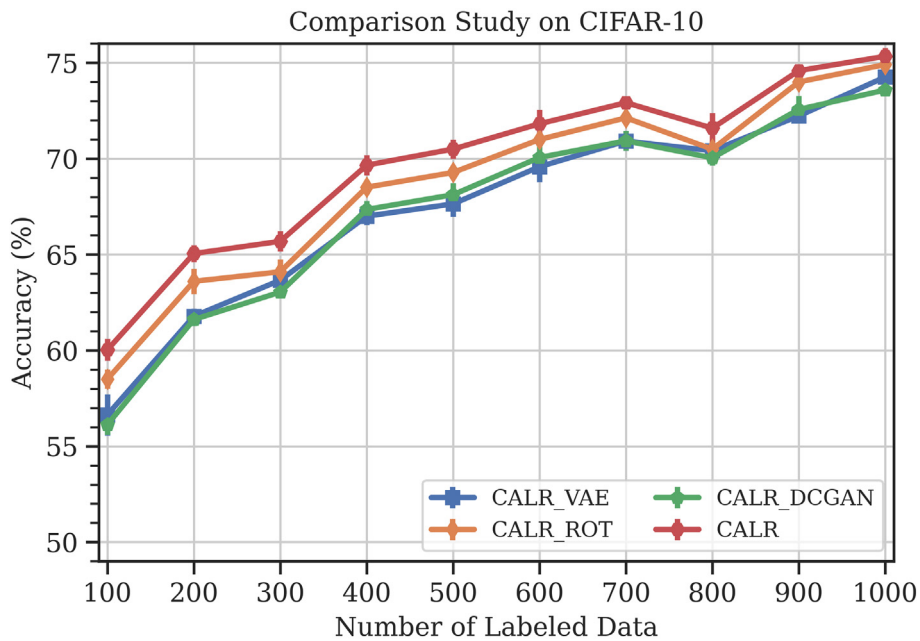


Fig. 12. Comparison experiments for feature representation on the CIFAR-10 dataset.

The results of these comparative studies on the CIFAR-10, CIFAR-100, and Caltech-256 datasets are shown in Figs. 15–17, respectively. CALR_Q4, the original version picking the maximum information density, performs best, while the performance of CALR_Q3 is the second best. In addition, CALR_RANDOM, the method using random sampling after BIRCH, and CALR_Q2, the method selecting the 50th sample of the information density, performed similarly on the three datasets. In addition, methods selecting samples with low information density, i.e., CALR_Q1 and CALR_Q0, have the worst performance. In summary, these experimental results demonstrate, to some extent, that the sample with the largest density of information can be typical of other samples in the cluster.

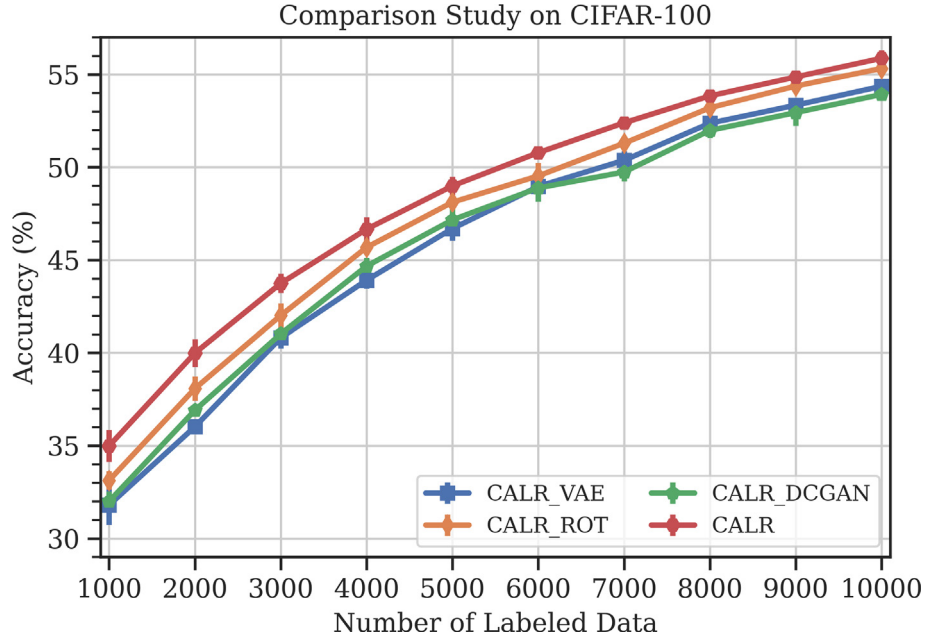


Fig. 13. Comparison experiments for feature representation on the CIFAR-100 dataset.

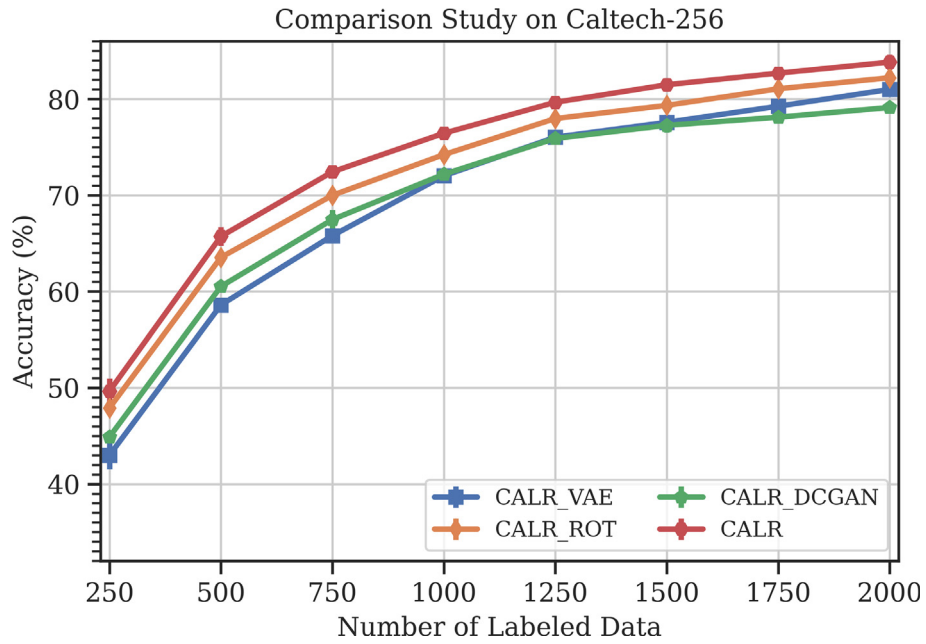


Fig. 14. Comparison experiments for feature representation on the Caltech-256 dataset.

4.9. Visualization of the sampling strategy

To fully understand the impact of our cold-start sampling model, we first demonstrate the properties of self-supervised feature extraction using dimensionality reduction visualization. Previous studies [46,47] showed that samples with similar high-level semantic attributes are more likely to be closer than samples with low-level appearances, e.g., black and white *T*-shirts would have more similar characteristics than black *T*-shirts and black birds. Furthermore, to verify this point in our

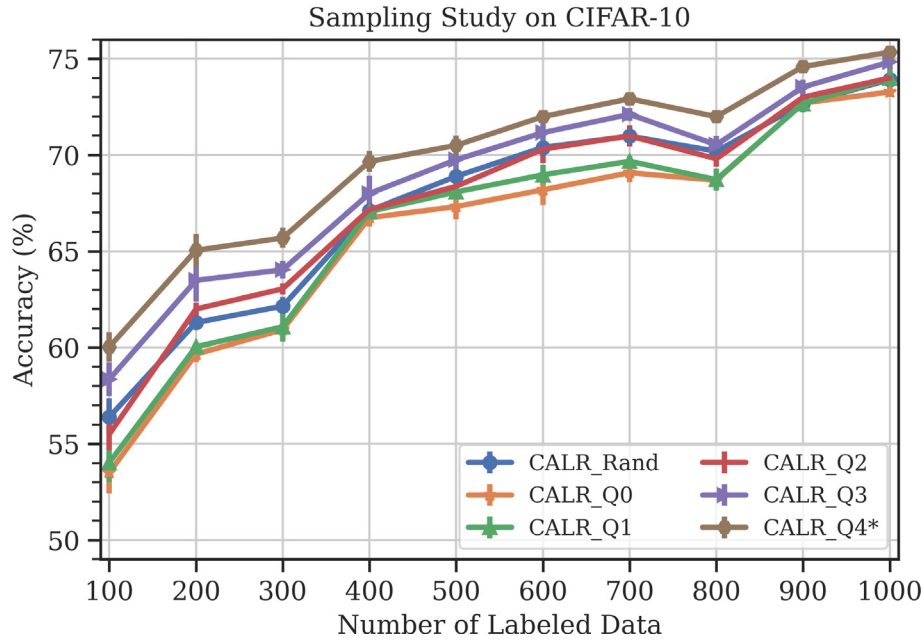


Fig. 15. Comparison experiments for different sampling methods on the CIFAR-10 dataset. * represents the sampling method used by our initially proposed framework.

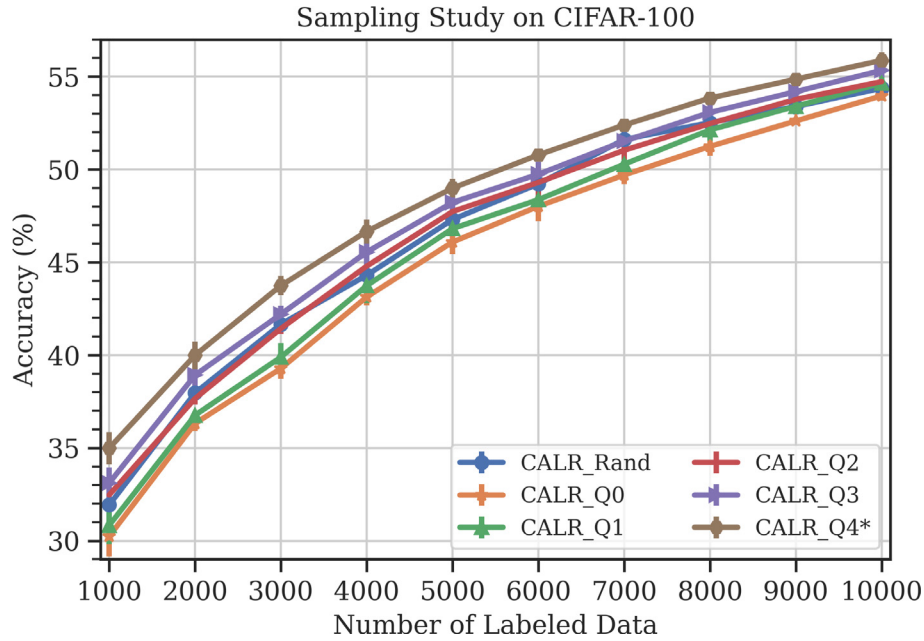


Fig. 16. Comparison experiments for different sampling methods on the CIFAR-100 dataset. * represents the sampling method used by our initially proposed framework.

case, we also carried out relevant T-SNE [48] visualization experiments. As shown in Fig. 18, there is a clear separation in the feature space between the machine classes (airplane, automobile, ship and truck) and the animal classes (bird, cat, deer, dog, frog and horse), which shows that self-supervised contrastive learning can distinguish well between high-level semantic information even without access to annotations.

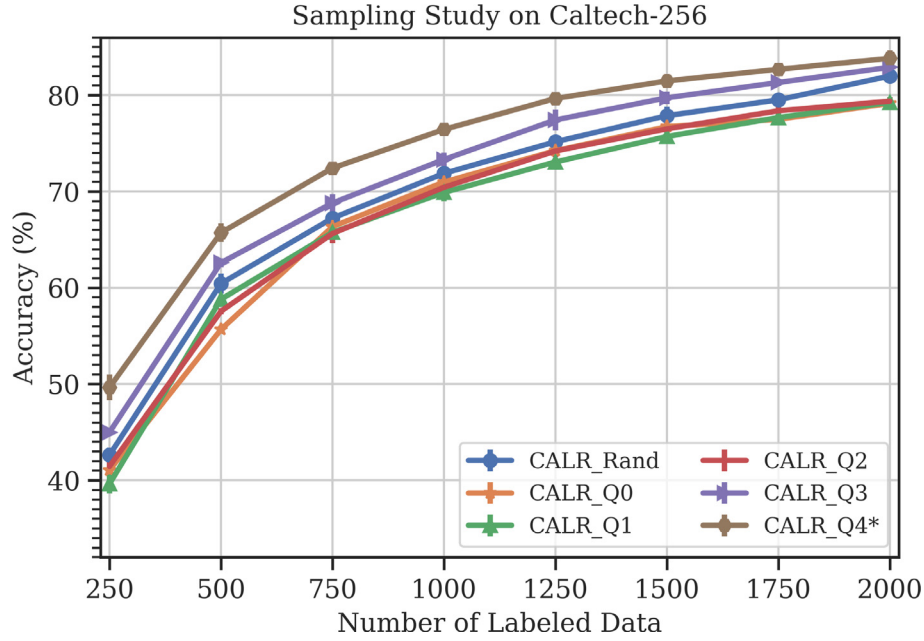


Fig. 17. Comparison experiments for different sampling methods on the Caltech-256 dataset. * represents the sampling method used by our initially proposed framework.

Second, we visualize the query preference of *CALR* for valuable samples in a two-dimensional Cartesian coordinate system. Specifically, we use scikit-learn [40] to artificially construct a category-balanced and a category-imbalanced toy dataset in Figs. 19 and 20. The four categories are represented by four colors: red, purple, brown and gray. We use two sampling methods, i.e., *Random* and *CALR*, to select 20 % of the samples. To increase the difficulty of sampling, we add Gaussian noise to make the decision boundary fuzzier.

For the category-balanced toy dataset, the visualization of sampling results is shown in Fig. 19. Compared with random sampling, our proposed method can better represent the distribution of the original samples, and its sampling results are similar to the ‘skeleton’ of the feature space. The BIRCH algorithm and maximum density sampling ensure the diversity and representativeness of the query results, respectively.

To further demonstrate the impact of class imbalances on AL, we visualize the query preference of *CALR* on toy data with a sample distribution of 100: 80: 40: 20 (red, purple, brown and gray color), as shown in Fig. 20 (a). It is easy to see the results in Fig. 20 that compared with random sampling, our proposed method can suppress the sampling of the most dominant class and make the category distribution of the sampling results more balanced. This is because we have used clustering preprocessing to ensure that the data of each category and each distribution can be selected.

5. Discussion

5.1. Transfer learning vs self-supervised learning

The experimental results show that self-supervised learning is more beneficial to cold-start AL. Compared with transfer learning from ImageNet, the self-supervised pretraining paradigm has no domain shift and can be used in more scenarios, such as medical image processing. Of course, if the target domain is close to ImageNet, transfer learning can also achieve good cold-start results. This is demonstrated in the ablation experiment on Caltech-256, i.e., *CALR_IN* could obtain suboptimal results second only to *CALR*.

5.2. Context-based feature representation vs generation-based feature representation

Based on the experimental results in Section 4.8, it can be found that context-based feature learning, e.g., contrastive learning and rotation recognition, is superior to generation-based feature learning, e.g., variational autoencoder and generative adversarial network. On the one hand, context-based feature learning mainly focuses on the semantic features of samples [49], such as context similarity and spatial structure. On the other hand, generation-based methods learn visual features through the process of image generation tasks, which focus more on underlying information [49], such as color, texture, and

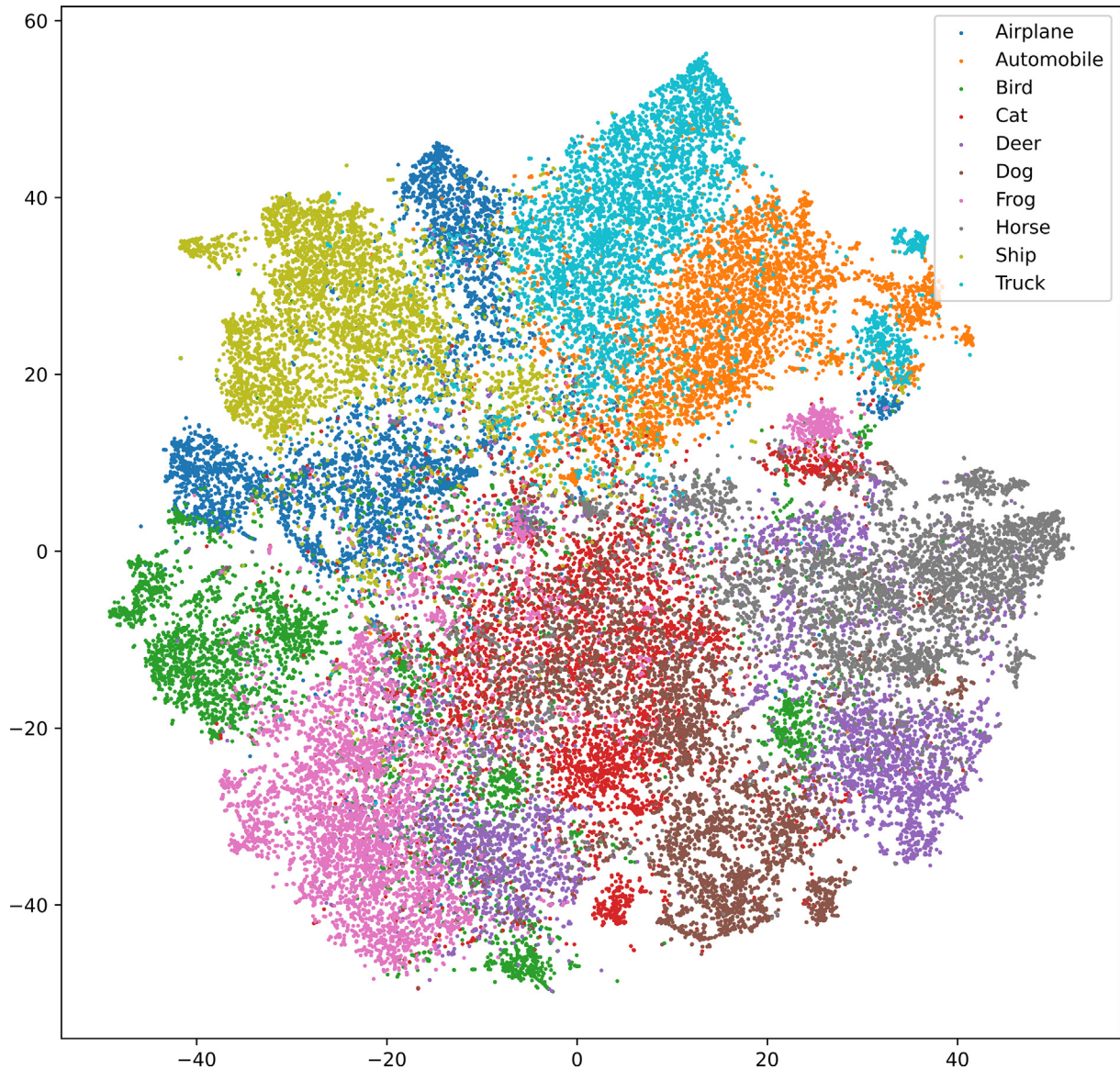


Fig. 18. Contrastive self-supervised feature T-SNE plot on the CIFAR-10 dataset. Each color represents one certain class in the CIFAR-10 dataset.

intensity. Therefore, we reason that context-based feature representation with more semantic information is more conducive to subsequent active selection.

5.3. BIRCH clustering vs K-means clustering

According to the results of ablation studies, we can easily notice that BIRCH clustering has better cold-start performance than k-means clustering. The reason might be that although the k-means clustering algorithm has been used widely, it is sensitive to outliers, which easily deteriorates the clustering performance. Moreover, k-means clustering may have some difficulties regarding the limited availability of samples and their imbalanced category distribution. Conversely, BIRCH can identify noise samples and is suitable for clustering tasks on imbalanced datasets.

5.4. Cold-start AL vs Warm-start AL

As a cold-start AL method, CALR is suitable for use when there is no initial annotation, the experts cannot be ready for annotation at any time, or the annotation budget and computing resources are limited. Experiments show that under any

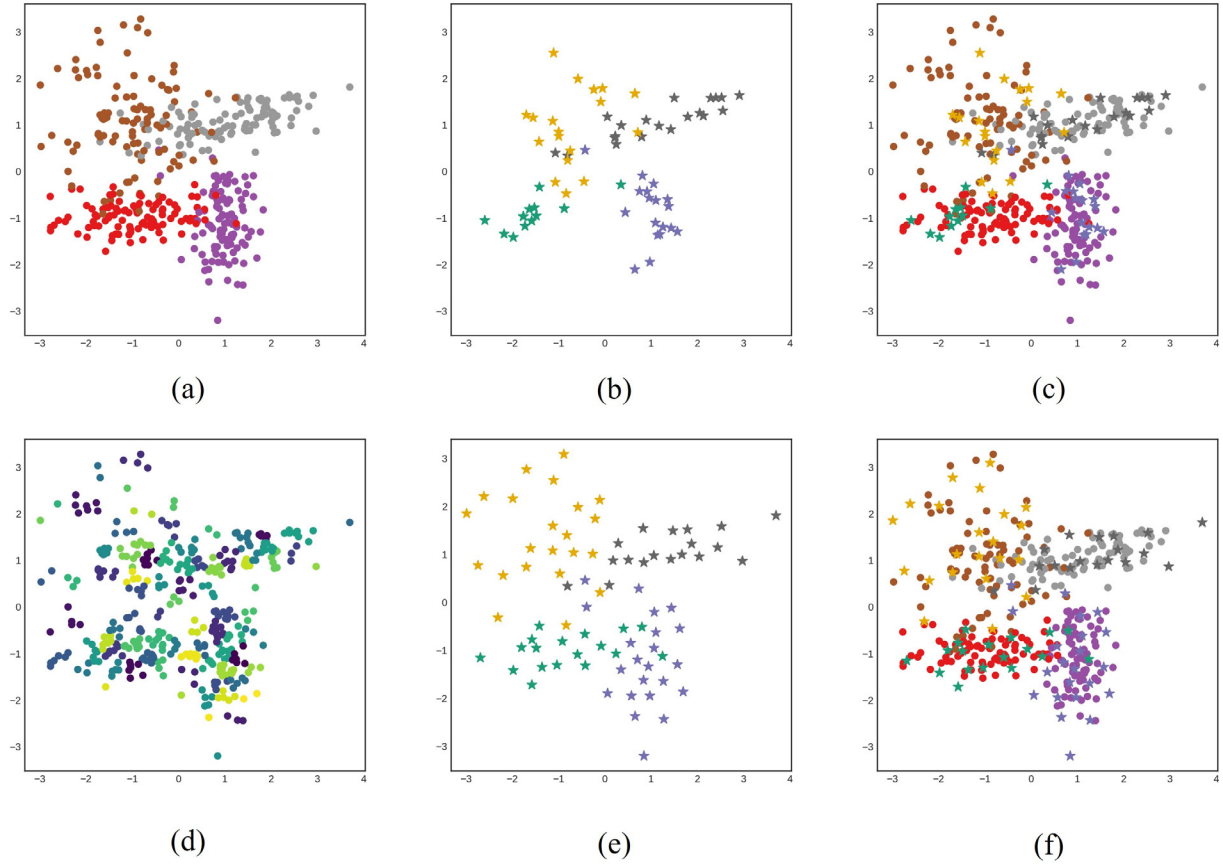


Fig. 19. Visualization of the query preference for sampling methods on a category-balanced dataset, where different colors represent different categories, and the pentagrams represent the selected samples. (a) is the original toy dataset, (b) is the selected samples of Random sampling, (c) is the visualization of the random sampling query results (including both selected and unselected samples), (d) is the clustering results, and different colors represent the different clusters, (e) is the selected samples of CALR, (f) is the visualization of the CALR query results (including both selected and unselected samples).

of these conditions, CALR can reduce annotation cost or help achieve better classification performance than any other method, including warm-start AL. In addition, for warm-start AL, CALR can also be used as one kind of initialization approach, and its performance is better than that of random initialization, which is demonstrated in Figs. 7 and 8.

5.5. Threats to validity (internal & external & constructive)

In this part, we will analyze threats to internal validity, external validity and constructive validity separately. More specifically, internal validity is the degree of confidence that the causal relationship we are testing is not influenced by other factors or variables. External validity is the extent to which our results can be generalized to other contexts. Constructive validity is about how well a test measures the concept it was designed to evaluate.

First, the main threat to internal validity is the budget for labeled data. That is, CALR can achieve cold-start AL efficiently and robustly under the condition of low budgets, while when the annotation data are sufficiently rich, perhaps combining CALR with other warm-start approaches would be a better option, which was analyzed in detail in Section 4.6.

Second, the main threat to external validity is the category of pattern recognition tasks. That is, CALR focuses on the specific pattern recognition task of classification, but its performance for segmentation, detection, registration and other tasks is not yet known and will be investigated in future works.

Third, the main threat to constructive validity is the randomness that accompanies the deep learning process. That is, for a convolutional neural network, it is possible that the results will be trapped in local optima, i.e., the classification performance is affected by the random seeds. Therefore, in our experimental design, we take different random number seeds and report their means and variances to avoid randomness affecting the reliability of the experimental results.

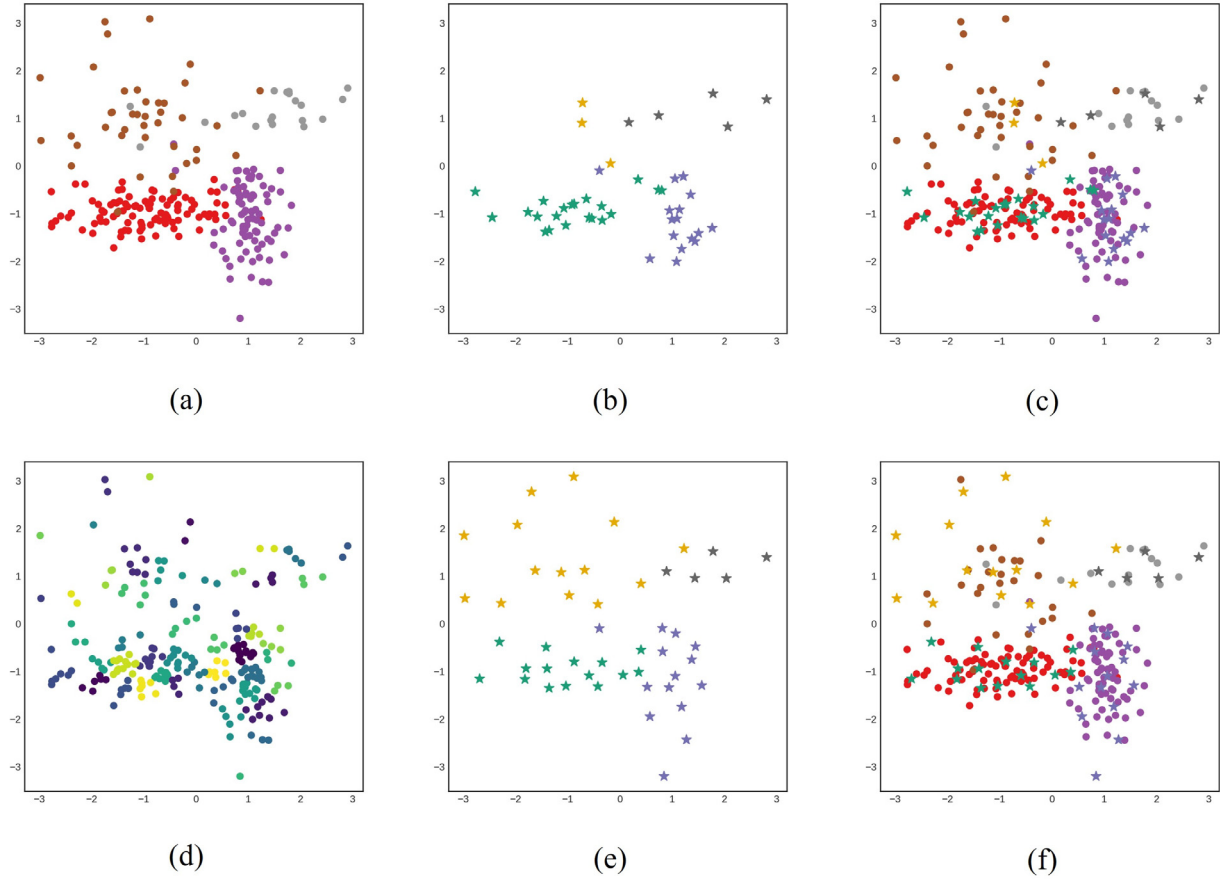


Fig. 20. Visualization of the query preference for sampling methods on a category-imbalanced toy dataset, where different colors represent different categories, and the pentagrams represent the selected samples. (a) is the original toy dataset, (b) is the selected sample of *Random* sampling, (c) is the visualization of random sampling query results (including both selected and unselected samples), (d) is the clustering results, and different colors represent different clusters, (e) is the selected sample of *CALR*, (f) is the visualization of *CALR* query results (including both selected and unselected samples).

6. Conclusion

We propose a cold-start active learning framework CALR, which consists of a contrastive self-supervised feature extractor, a hierarchical clustering module, and a sampling module based on information density. CALR aims to achieve cold-start AL efficiently and robustly under the condition of low budgets, low resources, and even imbalanced category distribution. We conducted classification experiments on the well-known CIFAR-10, CIFAR-100 and Caltech-256 datasets. The experimental results show that CALR can select label-efficient samples without using any initial labeled information and only needs to ask for queries from experts once. This work fills the research gap of cold-start AL in the field of computer vision.

In the future, we will study more advanced models, such as active learning methods capable of noisy oracle problems. That is, to determine what techniques can be used to mitigate the suspected degradation of annotation quality when using nonexpert oracles to avoid any potential loss in accuracy. In addition, the label ambiguity problem also arises in warm-start AL, and we will also investigate this kind of problem. That is, if the foreground objects of the pictures are small and the background accounts for a large proportion, how to guarantee the samples with the largest amount of information can be correctly selected.

CRedit authorship contribution statement

Qiuye Jin: Conceptualization, Methodology, Software, Visualization, Writing – original draft. **Mingzhi Yuan:** Data curation, Validation. **Shiman Li:** Writing – review & editing. **Haoran Wang:** Writing – review & editing. **Manning Wang:** Formal analysis, Supervision, Writing – review & editing. **Zhijian Song:** Investigation, Supervision, Resources, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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